UNIT 2

LECTURE 6

DEMAND-TYPES OF DEMAND

MEANING OF DEMAND

It refers to the quantity of a good or service that consumers are willing and able to purchase at various prices during a given period of time.

- Desire of a commodity
- Ability to pay for it
- Willing to pay for it
- At the given price
- At a given time

DEMAND SCHEDULE[2]

Demand schedule is that the tabular statement show various amount of a commodity which is demanded at different prices within a given period of your time.

Demand schedule is of 2 types

1. Individual demand schedule
2. Market demand schedule

INDIVIDUAL DEMAND SCHEDULE

Demand schedule is that the tabular statement show various amount of a commodity which is demanded at different prices within a given period of your time. Individual demand schedule may be a tabular statement showing the quantities of a given

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commodity which a consumer is willing to shop for in the least possible price within a given period of your time .

Individual demand schedule

Price of ice-cream (₹)	Quantity demanded by 'A' (units)
1	4
2	3
3	2
4	1

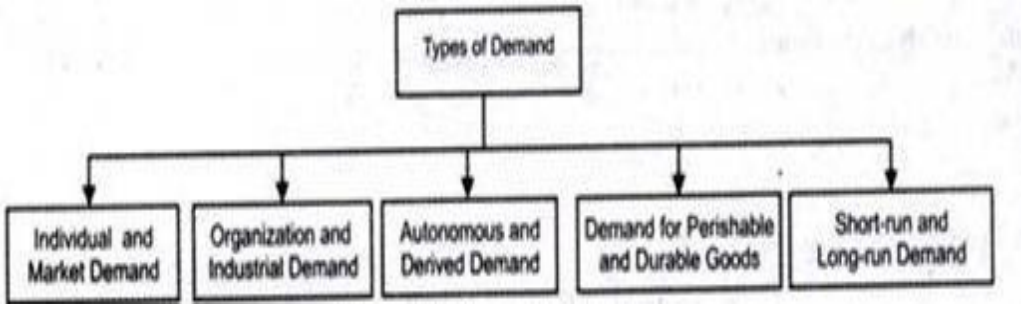
MARKET DEMAND SCHEDULE

Market demand schedule is a tabular representation showing the amount of a commodity demanded by all consumer in the market at all possible prices within a given period of time.

Market Demand Schedule

Price of A (per unit in pkr)	Individual Demand (per day)			Market Demand (per day)
	X	Y	Z	
4	1	3	4	8
3	3	4	5	12
2	4	5	6	15
1	5	9	9	23

Types of Demand[3]



The various types of demand are discussed as follows:

i. Individual and Market Demand:

Individual demand can be define as a quantity demanded by an person for a product at a particular price and within the specific period of time. For example, Mr. L demands 200 units of a product on Rs. 50 per unit in a week.

The individual demand of a product is partial by the price of a product, income of customers, and their tastes and preferences.

In simple language, market demand is the aggregate of individual demands of all the consumers of a product over a period of time at a specific price, while other factors are constant. on behalf of example, there are four consumers of oil (having a certain price). There are four consumers consume 30 liters, 40 liters, 50 liters, and 60 liters of oil respectively in a month. Hence, the market demand for oil is 180 liters in a month.

ii. Organization and Industry Demand:

The demand for the product of an organization at given price above a point of time is known as organization demand. On behalf of example, the demand for Toyota cars is organization demand. The total of demand for products of all organizations in a particular industry is known as industry demand.

For example, the demand for cars of various brands, like Toyota, Maruti Suzuki, Tata, and Hyundai, in India constitutes the industry' demand.

iii. Autonomous and Derived Demand:

The demand of a product that is not associate with the demand of other products is known as autonomous or direct demand. The autonomous demand arises due to the natural desire of an individual to consume the product.

On behalf of example, the demand for food, shelter, clothes, and vehicles is autonomous as it increases due to biological, physical, and other personal needs of consumers. On the other hand, derived demand refers to the demand for a product that increases due to the demand for other products.

On behalf of example, the demand for petrol, diesel, and other lubricants depends on the demand of vehicles.

iv. Demand for Perishable and Durable Goods:

The goods are separated into two categories, perishable goods and durable goods. Perishable or non-durable goods ask the products that have one use. For example, cement, coal, fuel, and eatables. On the opposite hand, durables ask goods which will be used repeatedly.

On behalf of example, clothes, shoes, machines, and buildings. Perishable goods satisfy this demand of people . However, durables satisfy both present also as future demand of people . Therefore, consumers purchase durable items by considering its durability.

v. Short-term and Long-term Demand:

Short-term demand means to the demand for products that are used for a shorter duration of time or for current period. This demand depends on the present tastes and preferences of consumers.

On behalf of example, demand for umbrellas, raincoats, sweaters, long boots is short term and seasonal in nature. On the opposite hand, long-term demand refers to the demand for products over a extended period of your time.

Generally, durable goods have long-term demand. The long-term demand of a product depends on variety of things , like change in technology, sort of competition, promotional activities, and availability of substitutes. The short-term and long-term concepts of demand are essential for an business to design a new product.

References:-

1. https://www.chegg.com/learn/economics/introductiontoeconomics/demand schedule.
2. https://www.economicsdiscussion.net/demand/5typesofdemandexplained/3355.
3. Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

LECTURE 7

DETERMINANTS OF DEMAND

Determinants of Demand[1]

A determinant of Demand means factor's that effects the demand.

Demand for a commodity or service is decided by variety of things. All such factors are called as demand determinants. Price of the given commodity, prices of other substitutes and/or complements, future expected trend in prices etc.

i. Price of the commodity

ii. Price of related commodities
- Complementary goods
- Competing/ substitute goods

iii. Income of the consumer
- Normal goods
- Inferior goods

iv. Tastes & preference of buyers(Demonstration effect)

v. Consumer's expectation

vi. Other factors
- Size of the population
- Composition of population
- Level of national income & its distribution
- Consumer credit facility & interest rates

Thus, several factors are responsible for bringing changes in the demand for a product in the market. A corporate executive should have the knowledge and knowledge about of these factors and forces so as to finalize his own production marketing and other business strategies.

1] Price of the Product

People use price as a parameter to form decisions if all other factors remain constant or equal. consistent with the law of demand, this suggests a rise in demand follows a discount in price and a decrease in demand follows a rise within the price of comparable goods.

The demand curve and the demand schedule help determine the demand quantity at a price level. An elastic demand implies a robust change quantity accompanied by a change in price. Similarly, an inelastic demand implies that volume doesn't change much even when there's a change in price.

2] Income of the Consumers

Rising incomes cause an increase within the number of products demanded by consumers. Similarly, a drop by income is amid reduced consumption levels. The link between income and demand is not linear in character. Marginal utility determine the proportion of change in the demand levels.

3] **Prices of related goods or services**

- **Complementary products** – An increase in the price of one product will cause a decrease in the quantity demanded of a complementary product. Example: Increase in the price of bread will reduce the demand for butter. This arises because the products are complementary in nature.

- **Substitute Product** – An increase in the price of one product will reason an increase in the demand for a substitute product. Example: Increase in price of tea will increase the demand for coffee and decrease the demand for tea.

4] Consumer Expectations

Expectations of a better income or expecting a rise in prices of products will cause a rise the number demanded. Similarly, expectations of a reduced income or a lowering in prices of goods will decrease the quantity demanded.

5] Number of Buyers in the Market

The number of buyer has a major outcome on the total or net demand. As the number increases, the demand rises. Besides, this is true irrespective of changes in the price of commodities.

6] **Taste and preference**

Taste and preference also affect demand of a commodity. Favorable taste results in increase in demand and unfavorable taste results in decrease in demand.

7] **Government policy**

Government policy also affect demand of a commodity. If government imposes tax then price of a product will increase and as a result demand will decrease.
And if government provide rebate then price of a product will decrease and demand of a product will increase.

8] **Distribution of income**

If there is equal distribution of income then there will be more demand for a product And if there's unequal distribution of income, then there'll be less demand for a product.

References: -

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2. https://www.scribd.com/document/425841371/ManagerialEconomicspdf
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LECTURE -8

DEMAND FUNCTION & ELASTICITY OF DEMAND

Low of Demand & its Limitations: -

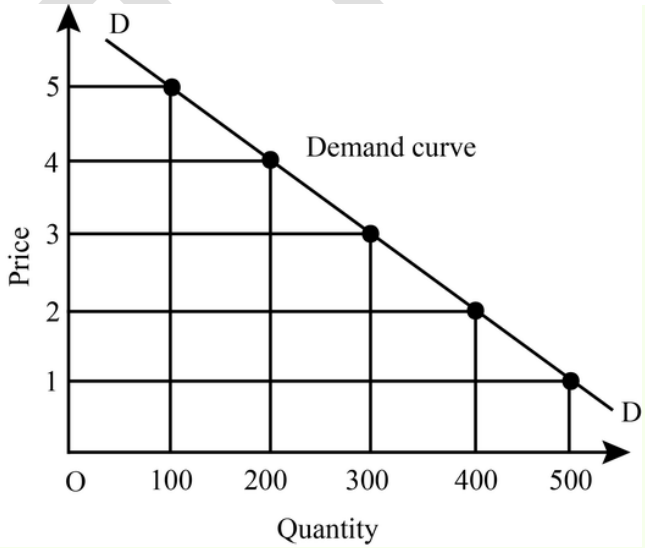
What Is the Law of Demand?

This law was first stated by Charles Davenant in 1699.

Law of demand states that there is an inverse relation between the price of a commodity and its quantity demanded, assuming all other factors affecting demand remain constant. It means that when the price of a good falls, the demand for the good rises and when price rises, the demand falls.

Law of demand may be explained with the help of the following demand schedule and demand curve: -

Demand Schedule	
Px	Dx
5	100
4	200
3	300
2	400
1	500



The above table and diagram show that as the price of the good reduces from Rs 5 to Rs 4, the demand for the good increases from 100 to 200 units.

Limitations of Law of Demand: -

When the prices of normal goods rise, the demand for them decrease, there are few cases where the law cannot operate. Following are the limitations of law of demand.

1. Prestige Goods

There are certain commodities like sports car or diamond, which are the sign of distinction and honor in any society. If the price of these goods increases the demand for them may be increase instead of falling.

2. Price Expectations

Expect a further increase in the price of a specific commodity they will go to buy more and more in spite of rising in price. In this case the violation of law is temporary. For Example Silver, Gold etc.

3. Ignorance of the Consumer

If the consumer is ignorant about the rise in price of goods, he may buy more of the commodity at higher price. For Examples:- Petrol, Diesel etc.

4. Giffen Goods

If the prices of basic goods like (sugar, wheat etc) on which the poor spend a large part of their income declining the poor increase the demand for superior goods. When the price of giffen goods fall it demand will also falls. There is a positive price effect in the case of giffen goods.

Meaning and Definition of Elasticity of Demand: -

Meaning and Definition of Elasticity of Demand

The term elasticity is borrowed from physics. It shows the response of one variable with value to a change in other variables on which it is dependent. Elasticity is an index of reaction.

In economics the term elasticity refers to a percentage of the relative changes in two quantities. It measures the reaction of one variable to the changes in another variable.

Definition: The **Elasticity of Demand** is a compute of change in the quantity demanded in reaction to the change in the price of the commodity. Simply, the effect of a change of price on the quantity demanded is called as the elasticity of demand.

Marshall, a renowned economist, has suggested a mathematical method to measure the elasticity of demand:-

$$E = \frac{\text{Percentage change in demand for a product}}{\text{Percentage change in the price of the product}}$$

According to this formula, the elasticity of demand can be define as a percentage change in demand as a outcome of the percentage change in price. Numerically, it can be written as:

$$E = \frac{\Delta Q}{\Delta P} X \frac{P}{Q}$$

Where,
$\Delta Q = Q_1 - Q_0$
$\Delta P = P_1 - P_0$
Q_1= New quantity
Q_2= Original quantity
P1 = New price
P0 = Original price

According to Marshall," The elasticity (or reaction) of demand in a market is great or small according to as the amount demanded much or little for a given fall in price, and diminish much or small for a given increase in price"

Types/Kinds of Elasticity of demand:-

3 Types of Elasticity of Demand

On the basis of different factors affecting the quantity demanded for a product, elasticity of demand is categorized into mainly three categories: **Price Elasticity of Demand (PED), Cross Elasticity of Demand (XED), and Income Elasticity of Demand (YED)**.

Let us look at them in detail and their examples.

1. Price Elasticity of Demand (PED)

Any change in the price of a commodity, whether it's a decrease or increase, affects the quantity demanded for a product. For example, when there is a rise in the prices of ceiling fans, the quantity demanded goes down.

This measure of responsiveness of quantity demanded when there is a change in price is termed as the Price Elasticity of Demand (PED).

The mathematical formula given to calculate the Price Elasticity of Demand is:

PED = % Change in Quantity Demanded / %Change in Price

2. Income Elasticity of Demand (YED)

The income levels of consumers play an important role in the quantity demanded for a product. This can be understood by looking at the difference in goods sold in the rural markets versus the goods sold in metro cities.

The formula given to calculate the Income Elasticity of Demand is given as:

YED *= % Change in Quantity Demanded / %Change in Income*

The result obtained from this formula helps to determine whether a good is a necessity good or a luxury good.

3. Cross Elasticity of Demand (XED)

In a market where there is an oligopoly, multiple players compete. Thus, the quantity demanded for a product does not only depend on itself but rather, there is an effect even when prices of other goods change.

Cross Elasticity of Demand, also represented as XED, is an economic concept that measures the sensitiveness of quantity demanded of one good (X) when there is a change in the price of another good (Y), and that's why it is also referred to as Cross-Price Elasticity of Demand.

The formula given to calculate the Cross Elasticity of Demand is given as:

XED ***= (% Change in Quantity Demanded for one good (X)) / (% Change in Price of another Good (Y))***

These three types of Elasticity of Demand measure the sensitivity of quantity demanded to a change in the price of the good, income of consumers buying the good, and the price of another good.

References: -

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2. Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

LECTURE 9

DEMAND FORECASTING – PURPOSE, DETERMINANTS AND METHODS

Demand Forecasting refers to the process of predict the future demand for the firm's product.

The art of predicting the demand for a service or product in the future market is known as demand forecasting. This forecast is based on historical data, past customer behavior trends, and patterns of the present market. In short, it is the scientific estimation of future demand for products and not just merely guessing it.

Importance of Demand Forecasting: -

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Determinants and Method of demand forecasting:-

some common demand forecasting methods, including:

1. **Statistical Method**

This approach uses historical data to identify patterns and trends that can be used to make predictions about future demand. This method is best suited for businesses with a long sales data history. It will be less effective for new businesses or have experienced significant changes in demand over time.

The statistical method can be divided into two sub-categories:

- **Trend Projection:** In this approach, demand is forecasted by extrapolating past demand data into the future. This method can be used when demand grows at a constant rate or when there is a clear trend in the data.
- **Regression Analysis:** This approach uses historical data to identify relationships between different factors and demand. The sourced information can be utilised to make predictions about demand in the future. This method is best suited for businesses with a large amount of data that can be analysed.

2. Market Research/Surveying Method

Before thinking about demand forecasting, you need to understand your target market. This means conducting extensive market research to get an idea of your target customers, their needs and wants, and what factors influence their buying decisions. One of the best ways to do this is through surveying. A survey can be either done by the company itself or by hiring a market research firm. Either way, surveys are a great way to get reliable data that you can use to make informed decisions about demand forecasting.

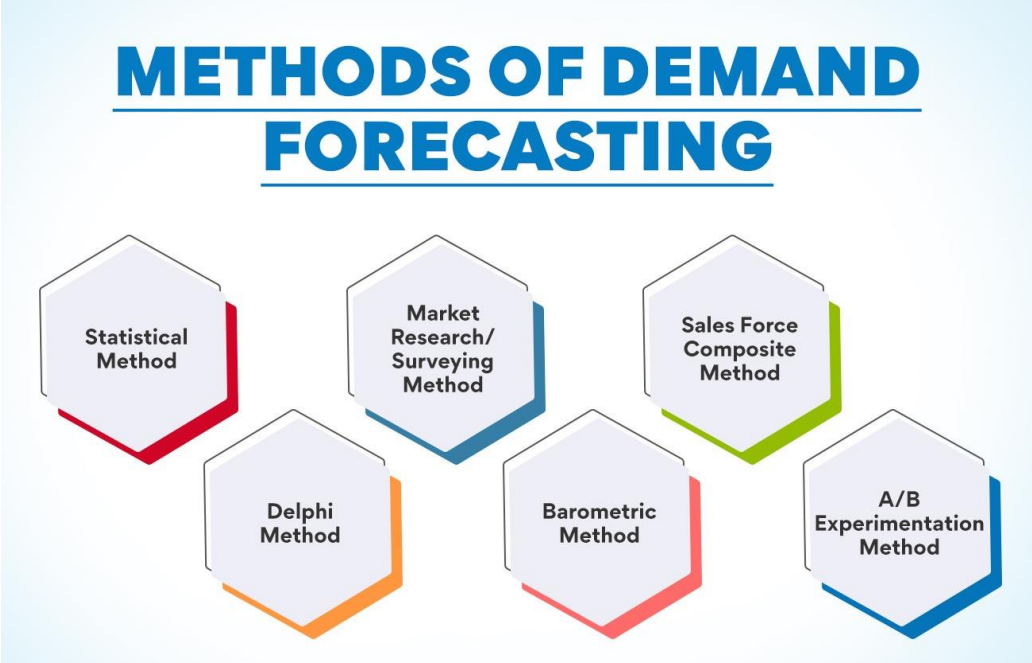
3. Sales Force Composite Method

The sales force composite method is a technique that relies on the collective knowledge of your sales team. Essentially, you ask your salespeople to estimate how much demand they think there will be for your product or service in the future. All the information is compiled and used to generate a demand forecast.

4. Delphi Method

It is a demand forecasting technique that relies on the opinions of experts. The Delphi method is used to generate forecasts by asking a panel of experts their opinions on the future demand for a product. The experts' responses are then collated, and an average demand forecast is generated.

The Delphi method is ideal when there is little historical data available on which to base a demand forecast. The Delphi method can be used to generate forecasts for both short-term and long-term demand.



5. Barometric Method

This demand forecasting method uses historical sales data to predict future demand. It's based on the theory that demand for a product is influenced by previous demand for that product. To forecast demand using this method, you'll need sales data for a similar product from past years. You can then use this data to identify patterns and trends to help predict future demand. The barometric method is best suited for products with stable demand. If demand for your product fluctuates significantly from year to year, this method may not be as accurate.

6. A/B Experimentation Method

This method is all about experimentation. You create two demand forecasts, one using your current methods and assumptions and a second demand forecast with different techniques and assumptions. Then, you compare the results of both demand forecasts to see which one is more accurate. It is an effective method when you have a limited amount of historical data to work with.

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LECTURE 10

SUPPLY-DETERMINANTS OF SUPPLY

What is Supply?

Supply is related to the price of the products, given that there is an incentive to supply at a higher price, as higher prices produce greater revenue and profits. Companies always want profit and, therefore, are more likely to manufacture more products at higher prices. When the price of goods or services is low, the supply is low, and when the price is high, the supply is also high.

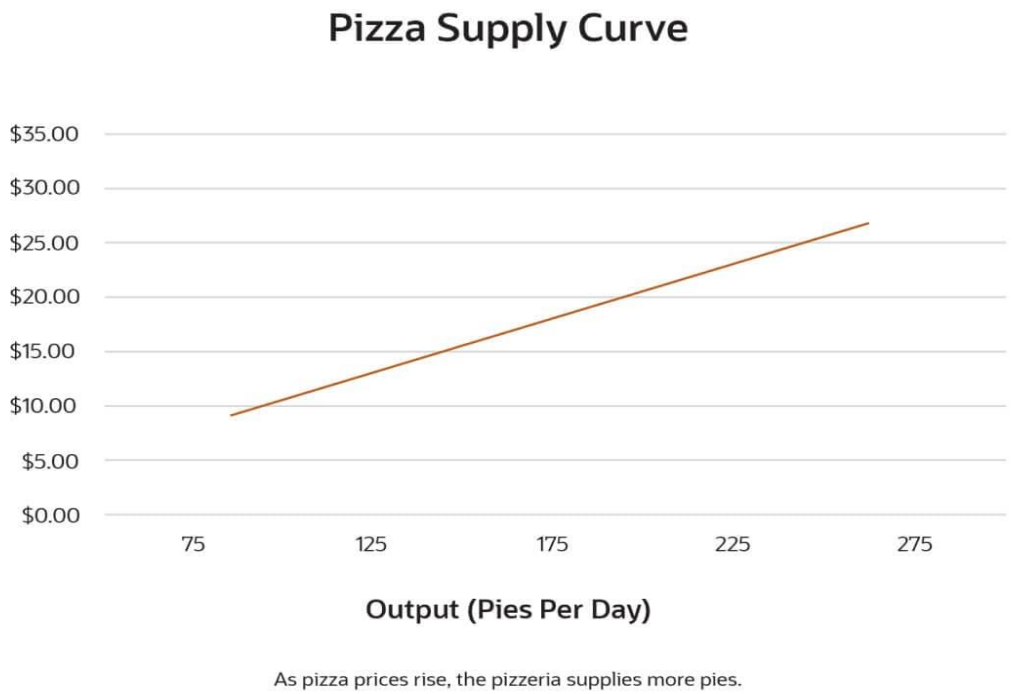
What Is the Law of Supply?

The law of supply is a basic economic concept. It states that an increase in the price of goods or services results in an increase in their supply. Supply is defined as the quantity of goods or services that suppliers are willing and able to provide to customers. The law works like this: Rising prices mean that products become more profitable, assuming other factors such as production costs remain constant.

Key Concept of Law of Supply: -

- The law of supply states that an increase in the price of goods or services results in an increase in the quantity that suppliers make available to the market.
- Existing suppliers increase production of higher-priced goods to maximize profits, while new suppliers may also enter the market.
- The law of supply assumes that all other factors remain constant. In practice, many other factors can play into supply decisions, including rising production costs and market competition.

- Consider the example of a pizzeria that sells pasta dishes as well as pizzas. If the price of pizza rises, and with it the profit per pie, the business may focus its resources on increasing the production of pizza — while decreasing the production of pasta offerings. As the price keeps rising, the pizzeria continues to increase the pizza supply because it can increase its profits by doing so. This relationship can be represented graphically as a supply curve, which shows the number of pizzas produced at different prices.



As pizza prices rise, the pizzeria supplies more pies.

References:-

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LECTURE 11
SUPPLY FUNCTION, ELASTICITY OF SUPPLY

Factors That Affect Supply: -

1. Price and demand forecasts: -

Many businesses base their production plans on forecasts of future demand and pricing, not just on what customers are currently buying.

2. Production costs: -

The law of supply assumes that companies can increase profits by selling more goods or services when prices rise, which provides them with an incentive to increase the supply. But if the price rises reflect increased production costs, that may not be true.

3. Competition:-

- New suppliers may enter the market even if prices are not increasing and demand is stable.

4. Technology:-Technology can enable companies to make and sell more products at a lower cost, thus increasing the available supply.

5. Transportation:-Transportation delays or rising shipping costs can affect a company's ability to increase its supply of goods. If goods can't move from warehouses to retail shelves, they can't be purchased by customers and don't count toward the market supply.

6. Availability of raw materials and labor:- A business may want to increase the supply of a product but unable to do so because it can't purchase the raw materials or hire the people required to produce it.

7. Government regulations and subsidies influence supply in some industries:- Companies must meet strict regulatory requirements when introducing certain healthcare products, for example, which can limit the supply of these products regardless of the demand. On the other hand, government subsidies support the supply of some local transportation services.

8. Weather and natural disasters:-For many agricultural goods, the weather has a major impact on supply. A dry season or flooding can greatly reduce crop yields.

9. Comparable goods:- A change in the supply of one good can affect the supply of other goods. For example, if the market price of corn increases, farmers may dedicate more land to growing corn. As a result, they use less land for growing squash, so the supply of squash decreases.

Exceptions to Law of Supply:-

1. Shift in business plan:- If a business is shifting its market focus and plans to cease production of some products, it may temporarily increase the supply of those products at a low price to eliminate any remaining stock and raw materials. A business may also use this approach as an emergency measure if it needs cash in a hurry.

2. Monopoly: -When there's only a single supplier of a good or service, the company may be able to increase or decrease its supply or pricing irrespective of external factors.

3. Competitive pricing:-In a highly competitive market, businesses may increase the supply of their products while reducing the price to capture market share.

4. Expiring or dated goods:- If perishable goods near their expiration date, a business may increase their supply early to try to recoup some of the production costs before the goods become unsellable.

5.One-of-a-kind goods:- Handmade art or other rare goods cannot be easily reproduced, so the supply cannot expand even if the price rises.

6. Inelastic supply:-For many goods, including agricultural products, it is difficult to quickly adjust the supply even if the price rises. It can take months or even years for crops to reach maturity and become available to customers. For example, an apple farmer who adds trees to their orchard won't be able to harvest the fruit for several years.

References: -

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